

1. Document Control

- System Name: Application
- Version: 1.0
- Document Type: Functional Requirements Document (FRD)
- Status: Draft

2. System Overview

- The Application ingests and processes customer and order CSV data into Delta tables.
- It performs data cleansing, calculates order totals, manages SCD Type 2 changes, joins and aggregates data, and executes all steps through Delta Live Tables.

3. User Roles

- Data Engineer: Configures, manages, and monitors the data pipeline.
- Data Analyst: Consumes aggregated and detailed tables for reporting.
- System Administrator: Maintains Delta Live Tables platform.

4. Functional Scope

- Ingestion of customer and order CSV data from specified file paths.
- Creation and management of the Delta tables for customer, order, ordersummary, and customeraggregatespend.
- Data cleansing to remove "Null"/Null values and duplicates.
- Computation of TotalAmount in order table.
- Join of customer and order data, with SCD Type 2 change management.
- Aggregation of customer spend by Name and Date.
- End-to-end pipeline built and orchestrated via Delta Live Tables.

5. Functional Requirements

FR-CORE-001 — Source Ingestion

- Description: Load customer and order source CSVs from specified file locations into their respective Delta tables.
- Preconditions: Availability of source CSV files at defined paths.
- Main Flow:

1. Detect CSV files at specified paths.
2. Read files and validate schema conformity.
3. Load data into Delta tables named customer and order.

- Alternate Flows: If file not found, log error and alert Data Engineer.
- Acceptance Criteria: All source rows ingested into corresponding Delta tables without schema mismatches.

FR-CORE-002 — Schemas Enforcement

- Description: Enforce required columns in customer and order Delta tables.
- Preconditions: Delta tables must exist.
- Main Flow:
 1. Check table schema for required columns: customer: CustId, Name, EmailId, Region; order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.
 2. Reject records not conforming to schema.
- Alternate Flows: If schema mismatch detected, log and quarantine records.
- Acceptance Criteria: Only conformant records are loaded into tables.

FR-CORE-003 — Computed Column Addition

- Description: Add TotalAmount column to order table as $\text{PricePerUnit} \times \text{Qty}$.
- Preconditions: order table loaded with PricePerUnit and Qty columns.
- Main Flow:
 1. For each order record, calculate TotalAmount.
 2. Append computed column to table.
- Alternate Flows: If PricePerUnit or Qty missing, set TotalAmount to null.
- Acceptance Criteria: All order rows have valid or null TotalAmount as per rule.

FR-CORE-004 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order tables.
- Preconditions: Delta tables loaded with source data.
- Main Flow:
 1. Scan for "Null"/Null string and null values in all columns.
 2. Remove any records with such values.
 3. Scan and remove duplicate rows based on all fields.
- Alternate Flows: If cleansing leaves table empty, log warning.
- Acceptance Criteria: No "Null"/Null strings or duplicate records remain in tables.

FR-CORE-005 — Create Target Table ordersummary

- Description: Create ordersummary table in defined catalog and schema if not exists.
- Preconditions: Catalog and schema must exist (gen_ai_poc_databrickscoe, sdlc_wizard).
- Main Flow:

1. Check for existence of ordersummary table.
2. Create table if not found, with required schema.

- Alternate Flows: If table exists, skip creation.
- Acceptance Criteria: ordersummary table exists with required columns.

FR-CORE-006 — Join and Load with SCD Type 2

- Description: Join customer and order on CustId to load ordersummary with SCD Type 2 change handling.
- Preconditions: customer and order tables available, SCD keys prepared.
- Main Flow:

1. Join customer and order on CustId.
2. Identify changes in customer attributes.
3. Apply SCD Type 2 logic: Inactivate old record, insert new as Active, update StartDate/EndDate.
4. Load joined data into ordersummary.

- Alternate Flows: If no changes detected, no SCD update required.
- Acceptance Criteria: ordersummary contains complete SCD history for each customer.

FR-CORE-007 — ordersummary Schema Population

- Description: Populate ordersummary table with specified fields.
- Preconditions: Join performed and SCD logic applied.
- Main Flow:

1. Assign fields: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.

- Alternate Flows: If any field missing post-join, set as null.
- Acceptance Criteria: ordersummary rows contain all specified columns.

FR-CORE-008 — SCD Type 2 Change Handling Rules

- Description: Set Inactive/Active flags, and manage StartDate and EndDate for SCD Type 2 tracking in ordersummary.
- Preconditions: SCD candidate changes detected.

- Main Flow:

1. Mark old customer records as Inactive, set EndDate.
2. Insert new customer records as Active, set StartDate.
3. Ensure only one Active record per customer at any point.

- Alternate Flows: No customer change — no SCD handling required.
- Acceptance Criteria: Full SCD Type 2 history is present and correct.

FR-CORE-009 — Create Aggregate Table customeraggregatespend

- Description: Create customeraggregatespend table if not exists.
- Preconditions: catalog=sdlc_wizard, schema=gen_ai_poc_databrickscoe.
- Main Flow:

1. Check if customeraggregatespend exists.
2. Create with columns Name, TotalAmount, Date if absent.

- Alternate Flows: If already exists, skip creation.
- Acceptance Criteria: Table exists and is ready for aggregation.

FR-CORE-010 — Aggregate Loading

- Description: Aggregate spend from ordersummary by Name and Date into customeraggregatespend.
- Preconditions: ordersummary table present and populated.
- Main Flow:

1. Group ordersummary rows by Name and Date.
2. Compute sum(TotalAmount) for each group.
3. Load results into customeraggregatespend.

- Alternate Flows: If no orders present for a Name/Date, no row loaded.
- Acceptance Criteria: customeraggregatespend holds correct daily aggregates.

FR-CORE-011 — Implementation Modality

- Description: Implement all pipeline steps using Delta Live Tables.
- Preconditions: Delta Live Tables environment is available.
- Main Flow:

1. Configure each transformation and load step as a Delta Live Table.

2. Set up dependencies and DAG to enforce processing order.

- Alternate Flows: If step fails, error handling/alert as per default platform.
- Acceptance Criteria: Full pipeline is automated in Delta Live Tables.

6. Business Rules

- $\text{TotalAmount} = \text{PricePerUnit} \times \text{Qty}$ for every order.
- Remove "Null"/Null literals and duplicates in customer and order data.
- Join customer and order on CustId.
- Apply SCD Type 2: mark previous as Inactive and new as Active, manage StartDate/EndDate, and fully track historical changes.
- customeraggregatespend: group rows by Name, Date; sum(TotalAmount).

7. Error Handling

- Log missing CSV files and alert Data Engineer.
- Quarantine records that fail schema validation.
- Log and skip rows with missing or invalid data during load and cleanse steps.
- Log SCD update failures and alert, but continue processing.
- Monitoring by Delta Live Tables default logging.

8. Dependencies

- Source CSVs for customer and order must be accessible before pipeline starts.
- Delta Live Tables must be available and configured.
- ordersummary must be created before customeraggregatespend aggregation.
- All operations assume correct file paths and table permissions.

9. Acceptance Criteria

- Ingestion completes with no missing or malformed data.
- No "Null"/Null literals or duplicates in customer or order tables post-load.
- TotalAmount accurately calculated in order records.
- ordersummary includes all expected fields with correct SCD Type 2 history.
- customeraggregatespend accurately reflects daily spend per Name.
- Entire data pipeline is implemented and executed using Delta Live Tables end-to-end.

10. Architecture & Data Flow Diagram (Graphviz)

